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Numerical Modeling of Condensate Buildup Extending to Well Drainage Region Using Effective Upstream Mobility

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Development of a computationally efficient numerical scheme to model condensate banking extending to drainage region is presented. A unified near wellbore modelling approach described uses effective upstream mobility to resolve condensate banking; non-Darcy flow; and capillary number dependence of relative permeability. This work will provide an understanding on how to accurately model and improve forecast of gas and/or condensate reservoir.

Potential drop in well drainage region across the grid block interfaces can be significant. This is true in particular for gas condensate reservoirs with low to moderate permeability (poor rocks), and in the vicinity of gas-condensate wells with high productivity indices consistent with pressure transient analysis. In gas condensate wells due to fracking, wells can have high productivity index, which results in relatively small drawdown. Due to poor rock connectivity in neighbouring cells, interface drawdown can be significant. Consequently, the use of conventional pseudo-pressure integral (limited to 1/5 of perforated grid block dimension) partially resolves pressure dependence of upstream mobility and/or upstream fluxes. One approach to reduce pressure dependence of upstream mobility would require the use of fine grids, i.e., (local) grid refinement. Nevertheless, grid refinement comes with significant added computational cost, this is impractical for large scale field simulations with thousands of wells. The method presented here involves identifying drainage region around wellbores, whereby to simulate condensate banking (extending to drainage region) effective upstream mobility is computed numerically. Further, in a gas-condensate system, as condensate drops out available pore spaces are reduced, and velocity of free gaseous phase increases, consequently non-Darcy flow effects and velocity dependence of relative permeability become more pronounced. Concept of pseudo-pressure integral evaluation is extended to drainage region, where non-Darcy flow effects and capillary number dependence of relative permeability both naturally fit.

Drainage pseudo-pressure replaces the traditional single point upstream mobility with an integrated form that more accurately predicts condensate banking in the high drawdown regimes around the wellbore, and is independent of well and grid geometry. Numerical results are validated against fine grid results, demonstrating accuracy and consistency of the approach. Drainage pseudo-pressure method is tested on coarse meshes and demonstrated to yield numerical results comparable with those obtained on (non-uniform) grids with local refinements, and comes with much less computational expense.

The new method improves accuracy of reservoir simulation, and compared to its counterpart local grid refinement method is computationally efficient. For selected test case(s), the new approach is demonstrated to be 4X faster, and this gain in computational performance is directly proportional to number of wells and/ or model size.

Introduction

In a gas-condensate system, production data for some gas-condensate wells have shown that well productivity is significantly reduced when wellbore bottom-hole-flowing-pressure (BHFP) drops below the saturation pressure of the in-place fluid (Gondouin et al., 1967; Fussell 1973). This is true even for lean gas condensate reservoirs where maximum liquid drop out in the deep reservoir is as low as 1% (Afidick et al., 1994). It is generally accepted that this reduction in productivity is due to accumulation of condensed liquid in the near wellbore region, and for low producing rate this could be attributed to accumulation of liquid in the tubing string or annulus (Fussell 1973). The liquid phase accumulates in the wellbore region, and form a ring which progressively impairs the gas deliverability. Calculation of gas-condensate well deliverability has been a long-standing problem without a simple solution (Fevang and Whitson 1995). The difficulty in modelling gas-condensate system is that the wells are typically in regions where the pressure gradients are the largest and current well models tend to under/over predict gas production in the presence of condensate drop-out. In gas-condensate wells, high-pressure gradients induce both large condensate saturation and high gas velocities in the near-wellbore regions. This is because accumulation of the condensate liquid close to wellbore leads to reduction of the effective flow area, increasing the gas velocity (rates) at the wellbore. Gas and condensate relative permeability can increase significantly by increasing the flow rate, this effect known as positive coupling (Whitson et al., 2003; Jamiolahmady et al., 2005). In the vicinity of gas-condensate wells subject to capillary number and interfacial tension (IFT) the relative permeabilities tends to be immiscible. In the reservoir, far from the wellbore, flow rates are low and Darcy's law remains valid. High flow rates in the perforated regions may lead to significant deviations from Darcy's law (Lombard et al., 2000). For gas condensate system induced by high flow rates, the non-Darcy flow coefficient and its related skin factor is more pronounced. Local (adaptive) grid refinement or very fine grid simulations can better represent the physics, nonetheless it comes with added computational expense, e.g. see (Mott 1999; Jamiolahmady et al., 2005, Manzoor et al. 2021).

Numerical model confirming that condensate blockage reduces deliverability of radial gas condensate well was first proposed by (Kniazeff and Naville 1965; Eilerts et al., 1965). Fetkovich (Fetkovich 1973) studied the use of rate and time dependent blockage skin to model condensate banking. First gas rate equation that uses pseudo-pressure integral to model condensate banking dates back to the work of O'Dell and Fussel (O'Dell 1967; Fussel 1973). Fevang and Whitson (Fevang and Whitson 1995 and 1997) presented pseudo-pressure method whereby pressure and saturations can be determined from producing well stream composition without an additional run. Nonetheless their approach involves a non-linear solve. Manzoor et al. 2018 has described a near well bore modelling approach to resolve pressure dependence of rock and fluid properties, and does not involve non-linear solve. Conventional pseudo-pressure is limited to perforated grid cell and resolves pressure dependence of upstream mobility partially only. Development of unified near wellbore modelling approach has been detailed elsewhere (Manzoor et al. 2021/2024). New drainage pseudo-pressure method presented here resolves pressure dependence of upstream mobility extending to drainage region and yields results which are comparable to those of fine grid results.

Potential (pressure) drop in well drainage region across the grid block interfaces can be significant. This is true in particular for gas condensate reservoirs with low to moderate permeability (poor rocks), and in the vicinity of gas-condensate wells with high productivity indices as demonstrated from an actual pressure transient evaluation (Figure-1). In gas condensate wells due to fracking specified with negative/stimulation skin, wells have high productivity index (WI_l), which results in relatively small drawdown. However, in the neighboring cells due to poor rocks, the interface drawdown (potential) can be significant. Consequently, the use of conventional pseudo-pressure integral which is limited to Peaceman's equivalent radius (1/5 of perforated grid block dimension) partially resolves pressure dependence of upstream mobility/fluxes. As mentioned above, one approach to reduce pressure dependence of upstream mobility would require the use of fine grids which can be limited to near wellbore (drainage) regions, i.e., local grid refinement. Nevertheless, grid refinement comes with significant added computational cost, and is impractical for large scale field simulations with thousands of wells. The method presented here involves identifying drainage region around wellbores, whereby to simulate condensate banking (extending to drainage region) effective upstream mobility is computed numerically. Drainage pseudo-pressure option replaces the traditional single point upstream mobility with an integrated form that more accurately predicts condensate banking in the high drawdown regimes around the wellbore, and is independent of well and grid geometry. Numerical results are validated against fine grid results, demonstrating accuracy and consistency of the approach. Drainage pseudo-pressure method is used on coarse (uniform) meshes and demonstrated to yield numerical results comparable with those obtained on (non-uniform) grids with local refinements. The new method improves accuracy of reservoir simulation, and compared to its counterpart local grid refinement method, it is computationally efficient. For small test case selected below, the new approach is 4X faster, and this gain in computational performance is directly proportional to number wells and/or model size.

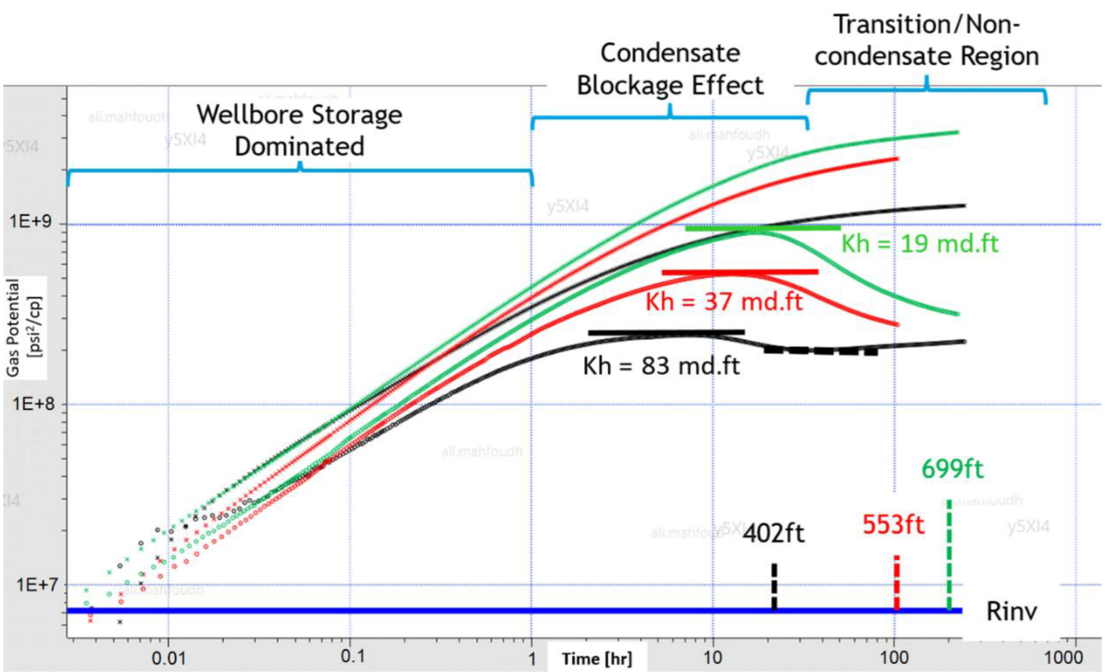


Figure 1—:Evolution of increasing condensate banking effect away from the well

Development

In a typical reservoir simulation, mathematical model comprised of a set of partial differential equations representing reservoir and well flows that are solved numerically. Numerical solution involves time and space/domain discretization replacing differential equations with difference equations. In each time step, a non-linear system is linearized using newton method and after discretization is solved iteratively, which may take several newton iterations to converge, e.g., (Coats 1980; A.H. Dogru et al. 2009). Domain subdivision also called grid generation involves subdividing reservoir domain into small grid blocks. A grid is a tessellation of a set of contiguous polygonal (2D) or polyhedral (3D) objects referred to as cells/elements/control-volumes. Numerical schemes used in reservoir simulation are control volume distributed (CVD), e.g. see (Manzoor et al. 2020). Rock properties such as permeability and porosity, and flow properties such as pressure, temperature, and composition (saturation) are assumed piecewise constant within a control volume. However, reservoir and flow properties may jump by order of magnitude across the faces of the control volumes. Consequently, property distribution in reservoir simulation is stair step, and rate of change of a property across grid blocks depends on grid resolution. Lack of definition within a single grid block and sharp changes in pressure and saturation across the grid blocks create several physical, numerical, and convergence problems.

In numerical simulation dynamic interaction between hydrocarbon reservoirs and wells has been an object of study for decades (C.K. Eilerts 1965; L.P. Dake 1978; A.U. Chaudhary 2003). Reservoir boundary conditions come in the form of well controls that are used to match historical data and/or define operational limits for reservoir forecasting. Figure 2, displays a perforated grid (corresponding to a representative producer well) together with its neighboring cell, with $ø_{G1}$ and $ø_{G2}$ are their respective grid block potentials. Here, $ø_G$ is interface potential, and can be calculated by inverting two-point flux approximation scheme (Edwards 1998). The potential drops caused by producing fluid from a perforated grid cell ($\Delta ø$) depends on flow and rock properties of reservoir and wellbore, and potential drop decreases monotonically as it moves away from perforated grid block. Phase potential ($\Delta ø_p$) for a phase p is given by:

$$\Delta Ø_p = \Delta P + \Delta P_{c,p} - \rho_p g \Delta h \tag{1}$$

where ΔP is the pressure drop or drawdown, $\Delta P_{(c,p)}$ is difference between phase capillary pressure, and $\rho_p g \Delta h$ defines pressure difference due to gravity head (Δh).

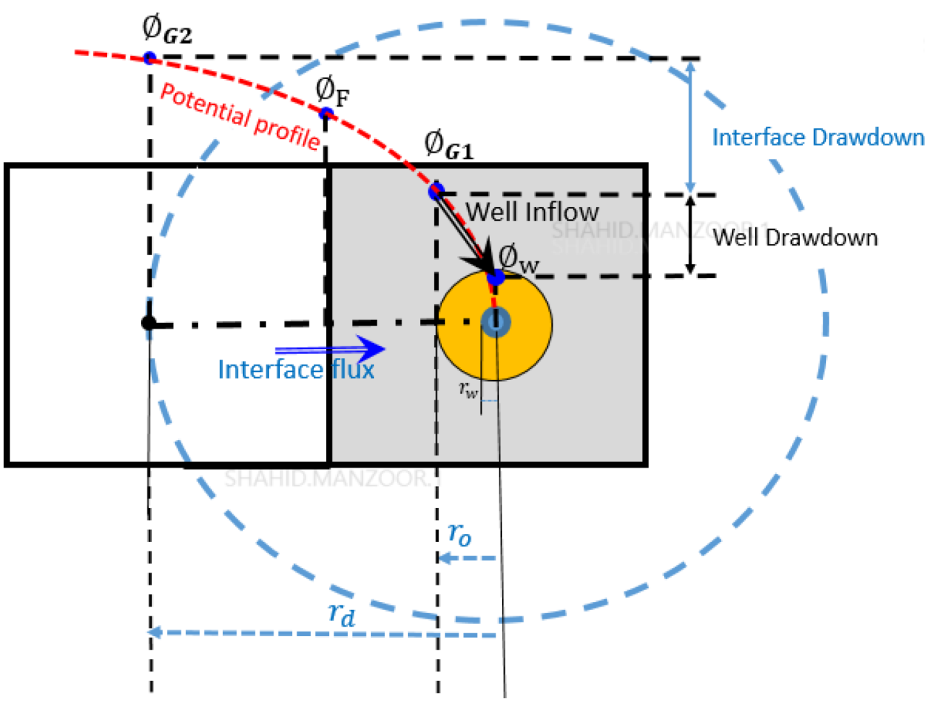


Figure 2—Schematic of well and interface drawdown and modeling condensate banking extending to wellbore drainage region.

In general, discrete grid cell size used in reservoir simulation is much larger than that of the wellbore (r_w) and would introduce singularities if the well was discretized similar to that of grid cell size. Well productivity index (WI_l) is introduced to couple wells to the reservoir, this relates wellbore pressure and flow to the grid cell parameters (D. Peaceman 1978). The well inflow performance relationship for a compositional case for a given phase p and component c through layer/completion l connected to grid block G_i, is given by:

$$q_{c,l} = WI_l \times \lambda_{c,l} \times \left(P_{Gi} - P_{w,l}\right) \tag{2}$$

where λ(c,l) is the upstream hydrocarbon component molar mobility, P_{Gi} is grid cell pressure, $P_{w,l}$ is the wellbore pressure incorporating gravity and friction effects for the layer l. For a hydrocarbon producing completion the mobility is given by:

$$\lambda_{c,l} = \left(\frac{k_{ro}\rho_o}{\mu_o}\right)_l x_c + \left(\frac{k_{rg}\rho_g}{\mu_g}\right)_l y_c \tag{3}$$

here k_{rp} represents relative permeability, ρ_p corresponds to molar density, and μ_p is viscosity with subscript p relating to hydrocarbon (oil/gas) phase. In Equation (3) x_c and y_c represent oil and gas component-mole-fraction respectively. The quantities defining mobility term ($\lambda_{c,l}$) are a function of saturation and pressure. In a reservoir simulator, wellbore and reservoir quantities are synchronized dynamically passing information from the wellbore to the reservoir and vice-versa. In general, molar mobility $\lambda_{c,l}$ is calculated using grid block quantities (pressure and saturations) and used in well inflow performance relationship, i.e., in equation (2).

Well-deliverability is a critical issue in the development of many gas-condensate reservoirs (Mott 1999; Fevang and Whitson 1995). In a gas-condensate system, production data for some gas-condensate wells have shown that well productivity is significantly reduced when wellbore flowing-bottom-hole-pressure (FBHP) drops below the saturation pressure of in-place fluid (Gondouin et al., 1967; Fussell 1973). This is true even for lean gas condensate reservoirs where maximum liquid drop out in the deep reservoir is as low as 1% (Afidick et al., 1994). The liquid phase accumulates in the wellbore region, forming a ring which progressively grows (extending to well drainage area) and impairs the gas deliverability. In the vicinity of gas-condensate wells where condensate saturations reach high values and oil relative permeability may exceed gas relative permeability, the gas dominated flow behavior is not at all correct, and use of upstream mobility leads to over prediction of field performance. One approach to reduce pressure dependence of upstream mobility would require the use of fine grids limited to near wellbore (drainage) regions, i.e., local grid refinement. Nevertheless, grid refinement comes with significant added computational cost, and is impractical for large scale field simulations with thousands of wells. Alternatively, to model condensate banking, multiphase transient wellbore model involving pseudo-pressure integral (limited to 1/5 of perforated cell dimension) is commonly used (Whitson et al., 1997; Jamiolahmady et al., 2005; Manzoor et al. 2018).

$$q_{c,l} = \beta \times WI_l \times \lambda_{c,l@P_{Gi}} \times \left(P_{Gi} - P_{w,l}\right) \tag{4}$$

Factor β is named as pseudo-pressure factor and is computed to resolve pressure dependence of fluid and rock properties, i.e., upstream mobility ($\lambda_{c,l@PGi}$) from wellbore radius (r_w) to Peacemen's equivalent radius (r_o).

$$\beta = \frac{\int_{P_{w,l}}^{P_{Gi}} \lambda_{TGMM} \mathrm{dp}}{\lambda_{TGMM@P_{Gi}} \times \left(P_{Gi} - P_{w,l}\right)}$$

where λ_{TGMM} is total generalized molar mobility given by:

$$\lambda_{TGMM} = \sum_{c=1}^{n\,comp} \left(\frac{k_{ro}\rho_o}{\mu_o} x_c + \frac{k_{rg}\rho_g}{\mu_g} y_c\right) = \frac{k_{ro}\rho_o}{\mu_o} + \frac{k_{rg}\rho_g}{\mu_g} \tag{5}$$

More details about pseudo-pressure integral evaluation can be found elsewhere (Manzoor et al. 2018). It is noted that the use of pseudo-pressure approach is limited to Peaceman's equivalent radius (r_o), which for uniform isotropic grid and properties is only 0.198 (1/5) times of perforated grid block size ΔX=ΔY, e.g., see Figure 2.

Potential (pressure) drop in drainage region across the grid block interfaces can be significant. This is true in particular for gas condensate reservoirs with low to moderate permeability (poor rocks) and gas-condensate wells with high productivity indices due to fracking involving negative/stimulation skin. In such cases due to negative skin well's productivity index (WI_l) is high, and consequently, well's drawdown (pressure drop) is relatively low in region-I (described below). Furthermore, in the neighboring cells due to poor rocks, the interface drawdown turns to be high. Therefore, the use of pseudo-pressure integral limited to region-I resolves pressure dependence of upstream mobility/flux partially. To date no physical/numerical model exist to model condensate banking extending to drainage region (II). In this work a novel approach is presented to model condensate banking extending to drainage region. In general, gas condensate flow and/or saturations are classified into three regions:

I. Near wellbore region where gas/oil flow is steady state and condensate build up is high. The near wellbore region is from wellbore radius (r_w) to Peaceman's equivalent radius (r_o).
II. Well drainage region is a second region, where pressure drop is relatively low and exhibits gas flow at a somewhat reduced permeability with condensate saturation being low but increasing in time.
III. Rest of the reservoir away from well drainage region, where pressure drop is negligible and assumption of upstream mobility is valid.

Potential drop across an interface is given by difference of potential between grid blocks sharing the interface, and is used in flux calculations. In a manner similar to well inflow performance relationship (i.e., equation 2) interface flux $\left(F_{\Gamma_{in}}\right)$ through an arbitrary interface Γ is given by:

$$F_{\Gamma_{in}} = -T \times \lambda_{\text{upstrm}} \times \left(\varnothing_{G1} - \varnothing_{G2}\right) \tag{6}$$

Where T is interface transmissibility, and λ_{upstrm} is upstream molar mobility, and in reservoir simulation is computed at upstream pressure (potential). We propose to extend the use of pseudo-pressure integral to drainage region thereby resolving pressure dependence of flow and rock properties. The method presented involves identifying drainage region around wellbores and extending pseudo-pressure integral to drainage region to resolve pressure dependence of upstream mobility and/or flux. Drainage pseudo-pressure option replaces the traditional single point upstream drainage mobility (Equation 6) with an integrated form that more accurately predicts condensate banking in the high drawdown regions around the wellbore, and is independent of well and grid geometry. The resulting modified interface influx relationship is then given by:

$$F_{\Gamma_{in}} = -\alpha \times T \times \lambda_{\text{upstrm}} \times \left(\varnothing_{G1} - \varnothing_{G2}\right) \tag{7}$$

Factor α is named as drainage-pseudo-pressure factor and is computed to resolve pressure dependence of fluid and rock properties, i.e., upstream mobility for all the cells within the drainage region. For a flux stream entering into an interface Γ_{in} with P_{upstrm}, and $P_{dwnstrm}$ being upstream and downstream pressures respectively, the factor a is given by:

$$\alpha = \frac{\int_{P_{dwnstrm}}^{P_{upstrm}} \lambda_{TGMM}\, dp}{\lambda_{TGMM@P_{upstrm}} \times \left(P_{upstrm} - P_{dwnstrm}\right)} \tag{8}$$

Note that P_{upstrm} is always less than $P_{dwnstrm}$. In Figure 2, for a producer well $P_{upstrm} = P_{G2}$ and $P_{dwnstrm} = P_{G1}$.

Implementation of method to compute effective upstream mobility: The procedure to implement drainage pseudo-pressure integral involves three key steps (Figure 3), which are described below. Constructing drainage region: To start with drainage pseudo-pressure integral, first step is to construct drainage region. In near wellbore region, cells defining drainage region are selected as candidate to resolve pressure dependence of upstream mobility. Starting with perforated cells, using cell adjacency list, and traversing cell interfaces, cells in the neighborhood of perforated cells are selected. In order to select drainage region, three different criteria can be used. For different well types, different criteria with different specification are supported. Further drainage region selection can be repeated subject to user specified frequency. These criteria are displayed in Figure 4, and detailed below.

- Cut off distances: How far to traverse from a perforated cell is controlled by user specified distances in X, Y, and Z directions.
- Cut off flux fraction (f): For a perforated cell, in general flux (moles drawn/produced) reduces as it moves away from the perforated cell. Let N_{well} be the well moles (well inflow), and $N_{interface}$ be the interface flux, then ratio of

$$f = \frac{N_{interface}}{N_{well}} < 1$$

  defines relative flux, and is used in selecting cells for drainage region. User can specify a minimum cut off relative flux, as a stopping criterion to limit selection of drainage region.
- Drainage Level: Another criterion supported to tag drainage region is based on level score with respect to perforated cells. Perforated cells are assumed with level-0, cells sharing interface(s) with the perforated cells define level-1. Further away, cells having an interface common with those of level-1 cells, defining level-2, and so on, e.g., Figure 4.

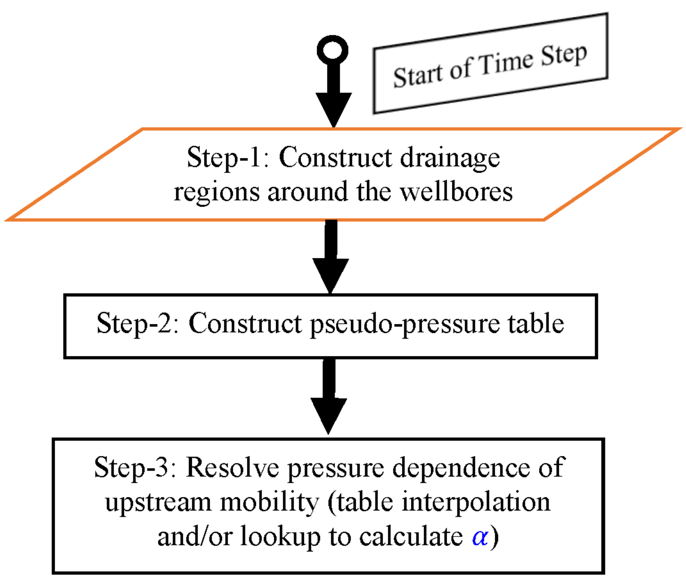


Figure 3—Flow chart highlighting key steps involved in method to compute effective upstream mobility

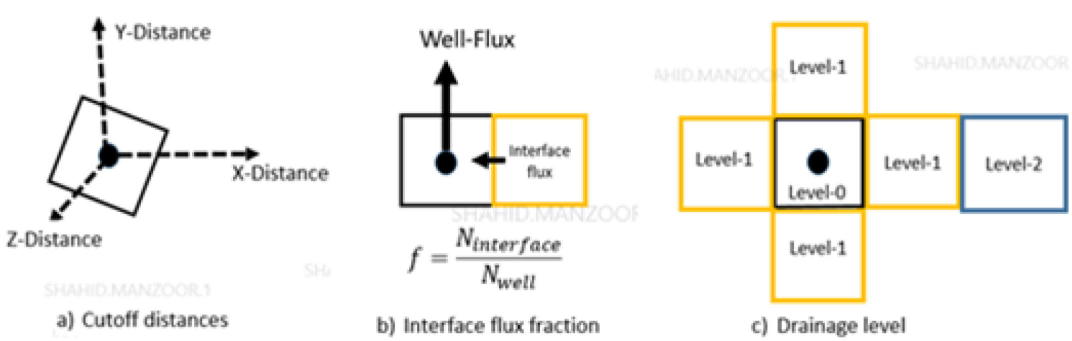


Figure 4—Criteria supported to select drainage region to resolve pressure dependence of upstream mobility

For an arbitrary test case by using above criteria, near well region selected to resolve pressure dependence of upstream mobility is displayed in Figure 5.

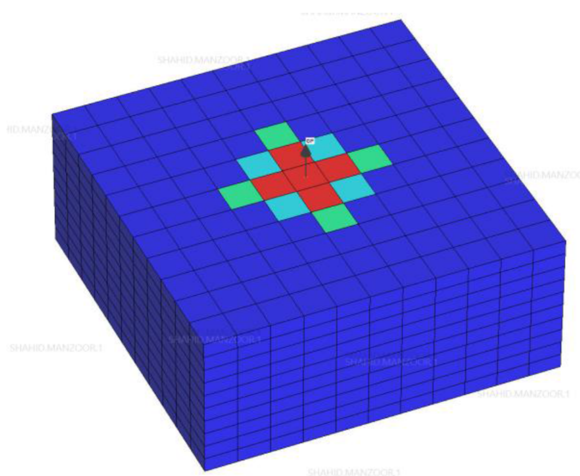


Figure 5—Example drainage region selected to resolve pressure dependence of upstream mobility

Construct pseudo-pressure table: For each cell with outgoing fluxes we propose to construct drainage pseudo-pressure table, e.g., Table-2. For a cell at hand, cell-center pressure defines lower limit of the drainage pseudo-pressure table. Each grid cell can have number of interfaces with outgoing fluxes. In order to construct a single pressure dependent mobility table, maximum of all neighboring cells pressure is used as upper pressure limit, and covers possible range of interpolation pressures. Constant composition experiment (CCE) is performed thereby constructing pressure versus total generalized molar mobility (λ_{TGMM}) table. In CCE, steady state conditions are assumed, with no net accumulation. This is achieved by enforcing that at any given pressure, ratio of hydrocarbon phase mobilities, and ratio of produced moles are same, e.g., below equation.

$$\frac{L}{V} = \frac{\lambda_o(S_o)}{\lambda_g(S_g)} = \frac{k_{rg}\rho_g / \mu_g}{k_{ro}\rho_o / \mu_o} \xrightarrow{yields} R = \frac{k_{ro}}{k_{rg}} = \frac{L \times \mu_o \times \rho_g}{V \times \mu_g \times \rho_o} \tag{9}$$

where V and L are the vapor and liquid hydrocarbon moles respectively obtained from CCE experiment. Since relative permeability is a function of saturation, and using constitutive relationship gas saturation (S_g) can be expressed in term of water (S_w) and oil saturations (S_o), i.e., S_g = 1- S_w+ S_o. Further, water saturation (S_w) is assumed at (upstream) grid block pressure (P_{Gi}) and is assumed constant in CCE. Equation (9) relates fluid and rock properties of hydrocarbons and is a non-linear equation in one variable, i.e., condensate (oil) saturation. Solving equation (9) iteratively can be computationally expensive, especially in full field simulations. We propose to construct a piecewise representation of rock-properties for a range of condensate saturations, eliminating the iterative solve. Piecewise representation of rock-properties yields a lookup table constructed by varying oil saturation S_o from a minimum (e.g.S_{or}) to maximum possible value (1-S_w), e.g. Table-1. Number of points in the proposed lookup table can be user specified and table can be adapted to relative permeability ratio to improve accuracy and performance. Table (1) displays such a piecewise representation of rock-properties. During CCE, for any integration pressures above dew point pressure with no drop out, k_{ro} is zero and iterative solve is not required.

Table 1—Piecewise representation of rock properties

S_o	k_{ro}	k_{rg}	k_{ro}/k_{rg}
0.243	0.000	0.737	0.000
0.278	0.001	0.644	0.002
0.312	0.008	0.541	0.016
0.347	0.028	0.432	0.065
0.382	0.076	0.325	0.233
0.417	0.192	0.226	0.847
0.451	0.498	0.142	3.510

Resolving pressure dependence of upstream mobility: The final result of above step is pressure versus λ_{TGMM} table, constructed for each selected grid block, covering range of interpolation pressures. A representative pressure versus λ_{TGMM} table is shown in Table-2. Note that drainage pseudo-pressure table is constructed at the start of every time step. Further, one table is constructed per selected cell, and is used to resolve pressure dependence for all outgoing interface fluxes. For each outgoing interface flux with known upstream and downstream pressures, drainage pseudo-pressure (blocking) factor α is calculated by integrating Equation (8) numerically. For numerical integration trapezoidal rule is used. The key steps involved in the proposed method are summarized in Figure 6.

Table 2—Representative drainage pseudo-pressure table constructed for a drainage cell

Pressure	ρ_g	μ_g	ρ_o	μ_o	$\frac{L \times \mu_o \times \rho_g}{V \times \mu_g \times \rho_o}$	k_{rg} (Interp.)	k_{ro} (Interp.)	λ_{TGMM}
P_{Gi}	0.00860	0.00029	0.00798	0.00148	0.1753370	0.16674	0.02928	5.0502
$P_{Gi} + \delta P$	0.00881	0.00030	0.00806	0.00147	0.1424098	0.17645	0.02510	5.2814
$P_{Gi} + 2\delta P$	0.00901	0.00031	0.00812	0.00147	0.1119425	0.18721	0.02098	5.5353
$P_{Gi} + 3\delta P$	0.00920	0.00032	0.00817	0.00150	0.0850407	0.19816	0.01681	5.7885
$P_{Gi} + 4\delta P$	0.00939	0.00033	0.00819	0.00155	0.0622850	0.20833	0.01293	6.0189
$P_{Gi} + 5\delta P$	0.00956	0.00034	0.00820	0.00161	0.0433870	0.21758	0.00948	6.2261
$P_{Gi} + 6\delta P$	0.00973	0.00035	0.00819	0.00168	0.0274237	0.22715	0.00623	6.4447
$P_{Gi} + 7\delta P$	0.00989	0.00035	0.00819	0.00176	0.0133494	0.23609	0.00318	6.6478
Phase transition across dew point pressure (below is dry gas)								
$P_{Gi} + 8\delta P$	0.01005	0.00036	0	0	0	1.00000	0	27.953
P_{Gj}	0.01020	0.00037	0	0	0	1.00000	0	27.903

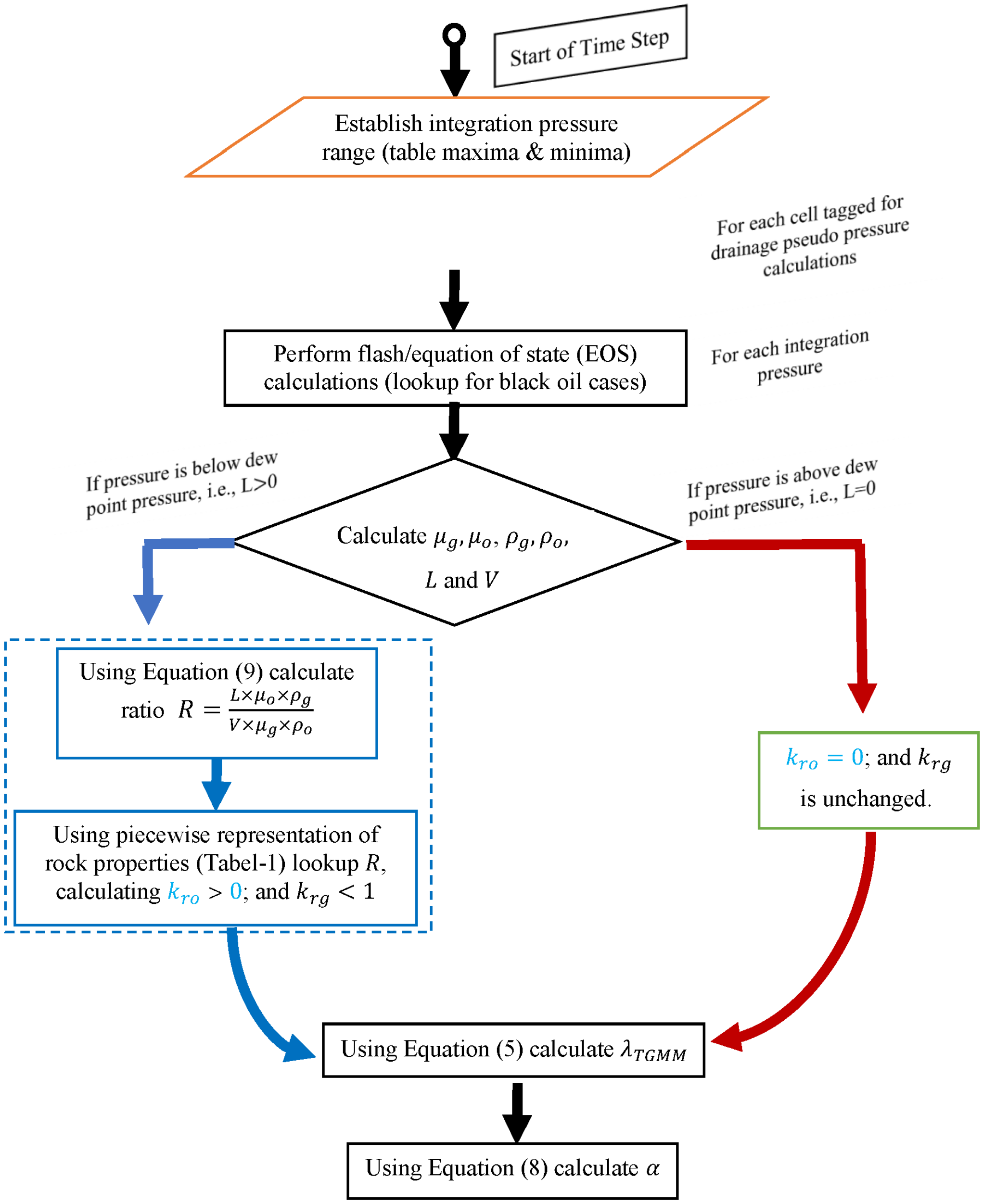
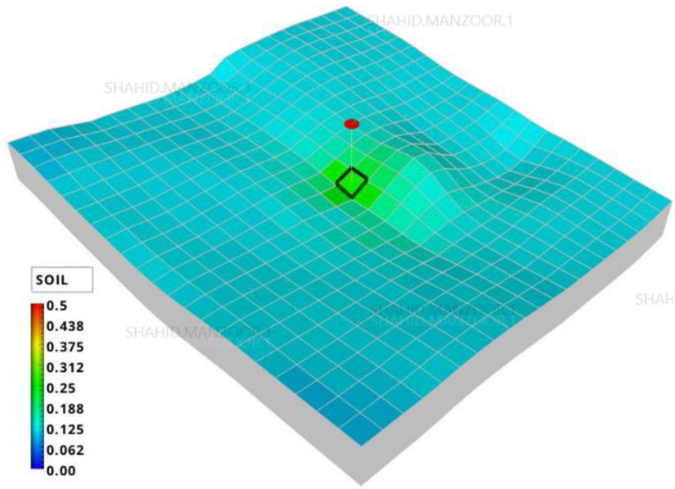


Figure 6—Flow chart highlighting key steps involved in drainage pseudo-pressure calculations

Numerical results

This test case is designed to validate development of drainage pseudo-pressure against fine grid results. Nine-components compositional test case is selected. Grid used is comprised of 21×21 cells and 106 layers, and is uniform (areal). Grid size in areal direction is 100m and layer thickness is 5*m*. The test case selected is heterogeneous, e.g., see condensate saturation map displayed in Figure-7. Rock properties are specified by oil-water and oil-gas saturation tables, with minimum residual saturations of water S_{orw} = 0.17 (S_{wc} = 0.05) and that of gas S_{org} = 0.12 (S_{gc} = 0.17) in their respective two-phase system. Stone-II model is used to interpolate three phase relative permeability from the two-phase data. Flow field is governed by a 5.0 *inch* diameter vertical well, which has perforations completed all the way from 41 to 89 layers. In each completed layer, skin=-2.65 is specified, this is to account for fracking and/or stimulation. Well is produced at 12000 *Mscf/day* and minimum flowing bottom hole pressure (BHP) is limited to 700 *psi*. Figure-7, shows condensate saturation map, and is clearly extending to well drainage region.



Property distribution	Minimum	Maximum	Average
Permeability: $k_x = k_y$	0.001mD	64mD	1.72mD
Porosity	0.00	0.24	0.0632

Figure 7—Uniform areal grid without local refinements

Figure 8, displays the simulation results without any pseudo-pressure and/or drainage-pseudo-pressure options being activated, well maintains specified rate until minimum BHP is reached. From Figure 8 it can be observed that there is condensate bank which is not accounted by the conventional well-model (Equation 2), and consequently on coarse grid (100m) use of upstream mobility leads to over predict performance of the well. This is verified by using local grid refinements and is discussed below.

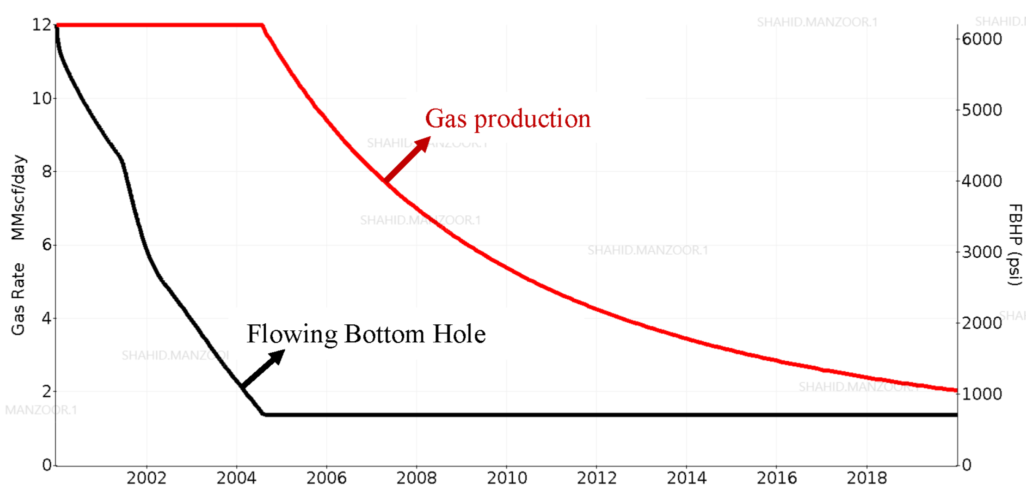


Figure 8—Gas production rate and flowing BHP plot simulated on uniform grid (Figure 7)

Now to reduce pressure dependence of rock and flow properties, local grid refinement with smooth transition is used. The grid employed together with condensate saturation map is displayed in Figure 9. Numerical results simulated using LGR are shown in Figure 10. Comparative performance shows that same test case when simulated using LGRs, is unable to maintain longer plateau, and well is constrained by minimum specified BHP. Note that in reservoir simulation regardless of coarse and/or fine scale mesh, upstream mobility is used. However, for coarse scale mesh with large size grid blocks due to high pressure drop across the grid block interfaces, pressure dependence of rock and flow properties is significant. On the other hand, for fine scale mesh pressure drop across the interfaces is relatively small and assumption of upstream mobility is valid and found to yield more accurate results, e.g., Figure 10 (solid versus dotted lines).

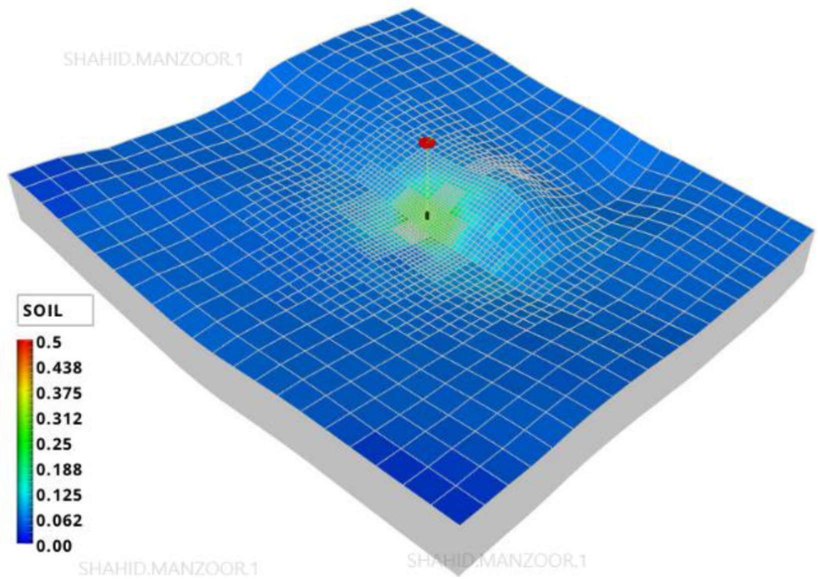


Figure 9—Example case with local grid refinement (smooth transition)

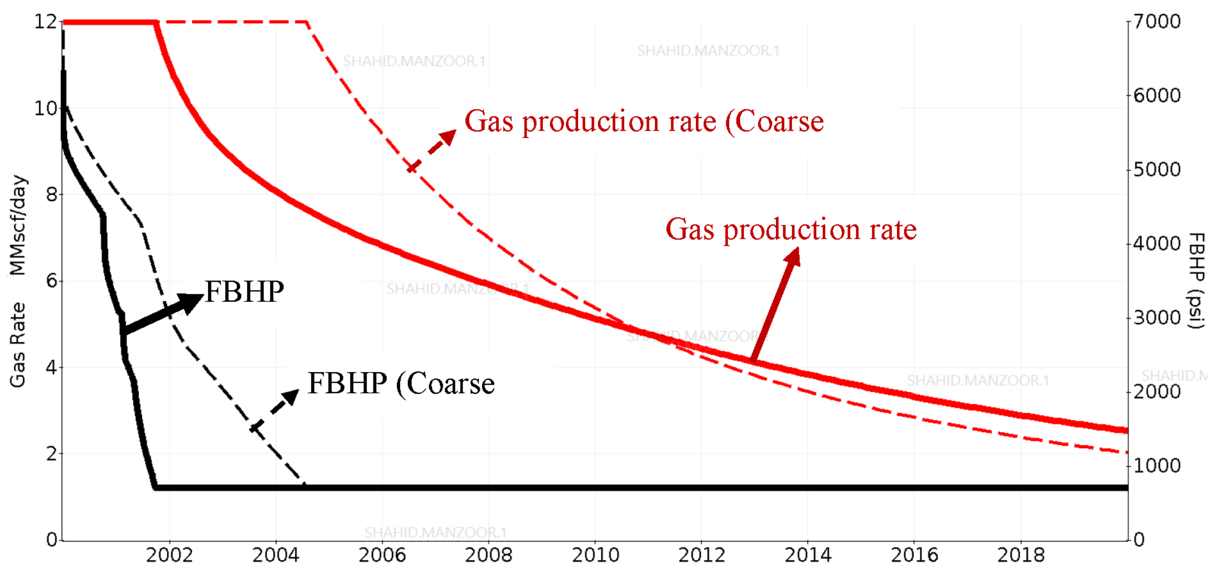


Figure 10—Comparative gas production rate and FBHP (LGR versus coarse grid)

In reservoir simulation use of LGRs in near well regions is known to improves the accuracy of numerical results. However, it comes with added computational cost and with increase in elapsed time. Even for single well studies, use of LGR can increase simulation time. Figure 13, displays comparative run time performance for the case at hand with LGR (60 min) versus without LGR (10 min). To assess comparative elapse time all the cases were run using same computer resources. For field scale simulation involving thousands of wells, use of LGRs, is associated with significant computational cost, and is impractical. Alternatively, a novel method drainage pseudo-pressure proposed here, which can be used on a coarse scale mesh without increasing grid resolution, and resolves pressure dependence of upstream flux extending to (drainage) region. Figure 11, shows cells tagged (red color) defines well drainage region.

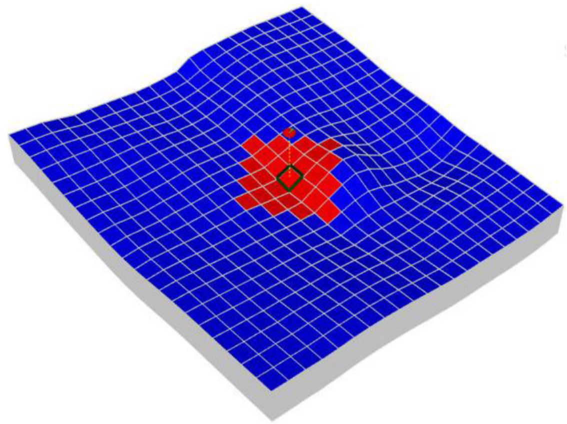


Figure 11—Cells tagged with red color define well drainage region

Next for each tagged cell we construct drainage pseudo-pressure Table. A presentative drainage pseudo-pressure table for a drainage cell is displayed in Table-2. For integration points with pressure below dew point, k_{rg} and k_{ro} are computed by interpolating R (given by Equation 9), to this end we use an adaptive s_o vs. k_{ro}/k_{rg} lookup table as shown in Table-1.

For discussion let us assume that Γ_{ij} be an interface with P_{Gi} and P_{Gj} be downstream and upstream pressures respectively. Table-2 be the drainage pseudo-pressure table. Note that across the phase transition λ_{TGMM} is significantly different. In conventional reservoir simulation upstream mobility is used (λ_{TGMM} = 27.903), and is significantly different than that of downstream mobility (λ_{TGMM} = 5.0502). In order to bridge this difference, we propose to use an effective/integrated upstream mobility thereby resolving pressure dependence of rock and flow properties. For the case at hand comparative gas production rate by using drainage pseudo pressure option is shown in Figure 12, which has been compared again fine grid results. Numerical result found to match closely with those of fine grid results. Note that drainage pseudo-pressure option does not require fine grid resolution, and compared to fine grid run, use of drainage pseudo-pressure option is computationally efficient, e.g., see Figure 13.

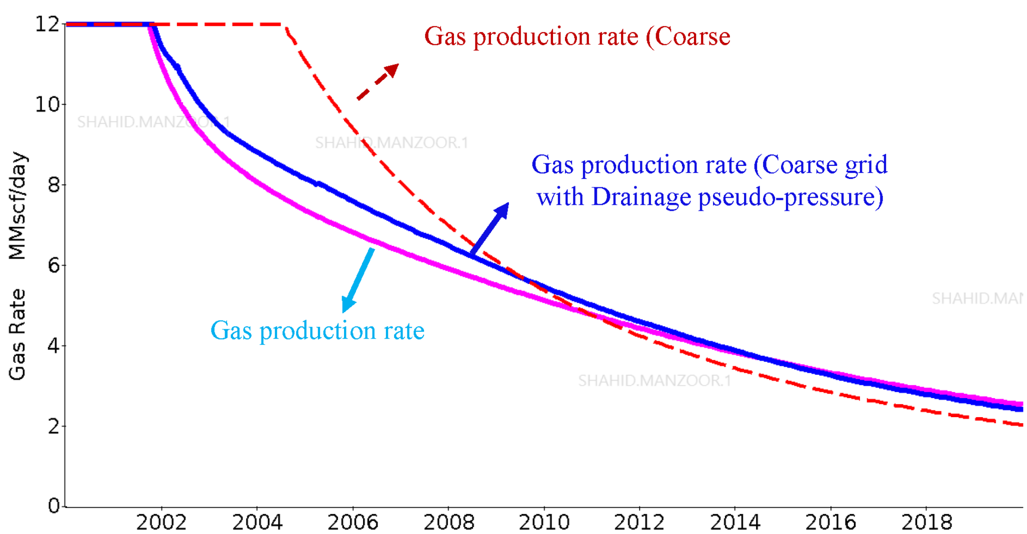


Figure 12—Comparative gas production rate: 1) with LGR; 2) without LGR; and 3) with drainage pseudo pressure applied on coarse scale mesh

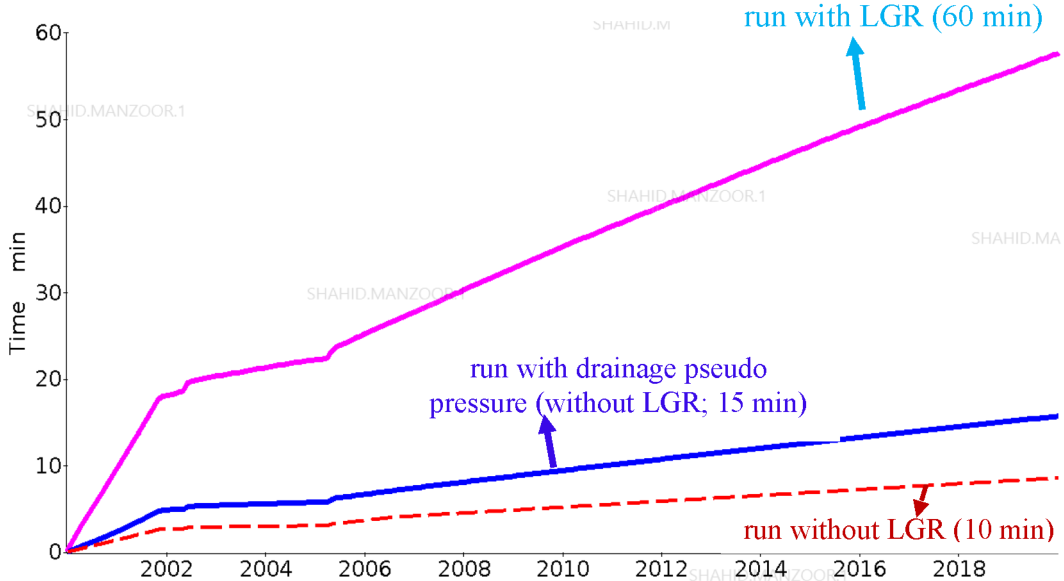


Figure 13—Comparative elapsed time: 1) with LGR; 2) without LGR; and 3) with drainage pseudo pressure applied on coarse scale mesh

For completeness we also study comparative performance of conventional pseudo-pressure versus drainage pseudo-pressure, e.g., see Figure 14. Conventional pseudo-pressure is limited to perforated grid cell and resolves pressure dependence of upstream mobility partially only. Whereas new drainage pseudo-pressure resolves pressure dependence of upstream mobility extending to drainage region and yields results which are comparable to those of fine grid results as shown in Figure 14.

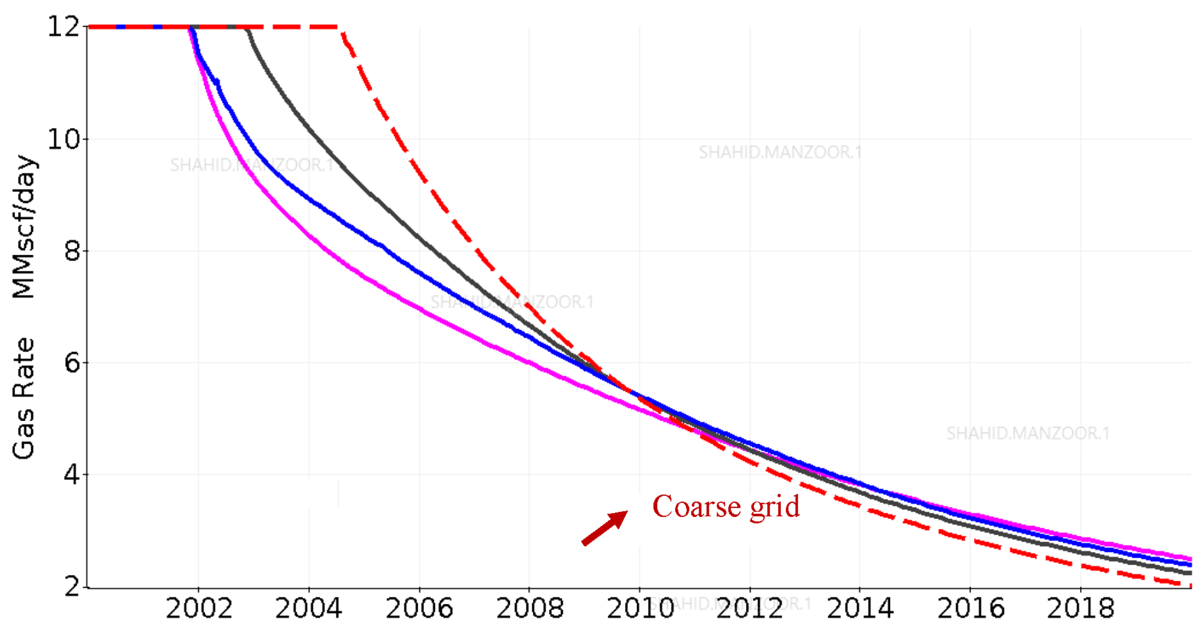


Figure 14—Comparative gas production rate conventional pseudo-pressure versus drainage pseudo-pressure.

Conclusions

In gas condensate wells due to fracking often specified with negative/stimulation skin, wells have high productivity index, which results in relatively small drawdown. However, in the neighboring cells due to poor rocks, the interface drawdown (potential drop) can be significant. Consequently, the use of conventional pseudo-pressure integral which is limited to Peaceman's equivalent radius (1/5 of perforated grid block dimension) partially resolves pressure dependence of upstream mobility/fluxes. One approach to reduce pressure dependence of upstream mobility would require the use of fine grids which can be limited to near wellbore (drainage) regions, i.e., local grid refinement. Nevertheless, grid refinement comes with significant added computational cost, and is impractical for large scale field simulations with thousands of wells. An alternative method presented here involves identifying drainage region around wellbores, whereby to simulate condensate banking (extending to drainage region) effective upstream mobility is computed numerically. Drainage pseudo-pressure option replaces the traditional single point upstream mobility with an integrated form that more accurately predicts condensate banking in the high drawdown regimes around the wellbore, and is independent of well and grid geometry. Numerical results are validated against fine grid results, demonstrating accuracy and consistency of the approach. Drainage pseudo-pressure method is used on coarse (uniform) meshes and demonstrated to yield numerical results comparable with those obtained by employing local grid refinements. The new method improves accuracy of reservoir simulation, and compared to its counterpart local grid refinement method, is computationally efficient. The computational performance gain is directly proportional to number wells and/or model size. The drainage pseudo-pressure method presented here can be unified with non-Darcy flow modelling and velocity dependent relative permeability options. For gas condensate reservoirs with low to moderate permeability pressure gradients/ drop in well drainage region across the grid block interfaces away from perforated grid cells can be significant. This behavior is consistent with pressure transient analysis. In such cases to resolve pressure dependence of rock and fluid properties use of drainage pseudo-pressure is recommended.

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